

SHOP DRAWINGS — ELECTRICAL / AGL SCOPE

RECTIFICATION OF TWY E BETWEEN E4 & E6 (ZIA)
AGL WORKS AT THREE (3) MILLING LOCATIONS — FIELD CONDITION SCOPE

Document No.	AUH-SK-AGL-TWYE-001 (sheets 1001 / 1002 / 1003) · Revision P08 — FINAL SCOPE OF WORK
Basis of scope	Final scope of work issued 30.07.2026 — supersedes the Rev P06 / P07 field-sheet quantities. Field verification sheets: Document_3 (LOC-01) · Second_milling_area rev _1 (LOC-02 / LOC-03).
Base layer	As-built AGL fitting positions (per assets_20260726120533.xlsx). AS PLOTTED they are ZIA LOCAL PROJECT GRID (m), matching the ADA source drawing to sub-millimetre — NOT UTM 40N. The Rev P05 EPSG:32640 label was incorrect — see sheet 6.
Prepared by	Mohammed — AGL Team Leader, ADB SAFEGATE AGL Team · Issued 28.07.2026 · Rev P08 30.07.2026

REV P08 — the final scope of work. All secondary routes at all three locations are SAW CUT; no duct route remains in the AGL scope. Every affected fitting takes a new SIDE-ENTRY SHALLOW BASE — 8" core at Location 1, 12" at Locations 2 and 3. The affected-asset list reduces from 30 to 16. A saw cut and side-entry base detail sheet is added and is held for the detail drawing. The as-built duct geometry, registration and stated limits of Rev P06 are carried forward unchanged.

SCOPE OF WORK & SEQUENCE — TWY E (E4–E6) · REV P08 (FINAL SCOPE)

SCOPE OF WORK — SEQUENCE OF AGL ACTIVITIES

PRE-ASPHALT — AGL ATTENDANCE

A1 Where secondary duct or secondary cable is laid by the civil team, a mandrel test is to be carried out on every duct and witnessed by the AGL team BEFORE asphalt laying. No asphalt to be laid over an unwitnessed duct.

POST-ASPHALT — AGL

- AGL installation works to commence only after written confirmation from the civil team that the asphalt curing period is complete.
- Coring after setting-out coordinates are issued by the civil survey team — 8" at Location 1 and 12" at Locations 2 and 3. Core diameter per the sheet schedule; do not assume one diameter throughout.
- Saw cutting to commence only after the survey points are confirmed in the field by the civil survey team, and not before the saw cut detail drawing is issued. At Rev P08 EVERY secondary route at all three locations is saw cut — no duct route remains in the AGL scope.
- Installation of new bases — SIDE-ENTRY SHALLOW BASE throughout: 8" at Location 1, 12" at Locations 2 and 3. At Location 2 the existing shallow base is cored out first.
- Laying of new secondary cable through the saw cut. No joints — full manhole-to-light replacement.
- Secondary cable termination, insulation resistance and continuity testing.
- Testing and commissioning of all affected circuits.
- Final functionality check, then handover to Operations for final inspection and return of the area to operational service.

HOLD POINTS & WITNESS REQUIREMENTS

H2 Curing complete — Civil. Written confirmation of the asphalt curing period before AGL works start (Item 1).

H3 Setting-out — Civil survey. Coordinates issued and field points confirmed before coring (Item 2) and before saw cutting (Item 3).

H4 Mandrel test — AGL witness. Every duct laid by the civil team, before asphalt laying (Item A1).

H5 Functionality check — AGL / Operations. Witnessed before handover (Item 8).

H6 Saw cut detail — AGL / Engineer. No saw cutting before the saw cut and side-entry base detail drawing is issued and accepted (Rev P08).

TECHNICAL QUERY — CONFIRMATION REQUIRED (NOT A CONSTRUCTION INSTRUCTION)

Subject: existing 4 x 110 mm secondary duct — cover, reuse and continuity

Field condition: the existing 4 x 110 mm secondary duct bank was exposed at 50 mm milling depth.

Q1. Confirmation required from the civil team before any duct is re-laid on the existing duct alignment.

Q2. The existing duct cannot be reused for the new secondary cable. The existing cables are obsolete and are to be withdrawn, and the existing duct cross-section will not accommodate the new secondary cable.

Q3. If new duct sized for the new secondary cable is laid within the construction area, confirmation is required from Civil on how the transition into the duct in the non-construction area is to be made. The same condition is expected over the full length, which would otherwise require saw cutting for the full length.

AGL TEAM PROPOSAL

Lay a full-stretch new secondary duct sized to accommodate the new secondary cable, giving a continuous route end to end and minimising saw cutting in the finished pavement.

Status at Rev P08: NOT ADOPTED. The final scope of work issued 30.07.2026 adopts SAW CUT for every secondary route at all three locations, which answers Q3 — the full length is saw cut and no new duct is laid under the AGL scope.

Q1 and Q2 remain OPEN on the existing 4 x 110 mm duct bank: whether it is to be left in place, and confirmation that the obsolete cables are withdrawn. To be raised under RFI AUH-WP-AGL-TWYE-001. Not a construction instruction until answered.

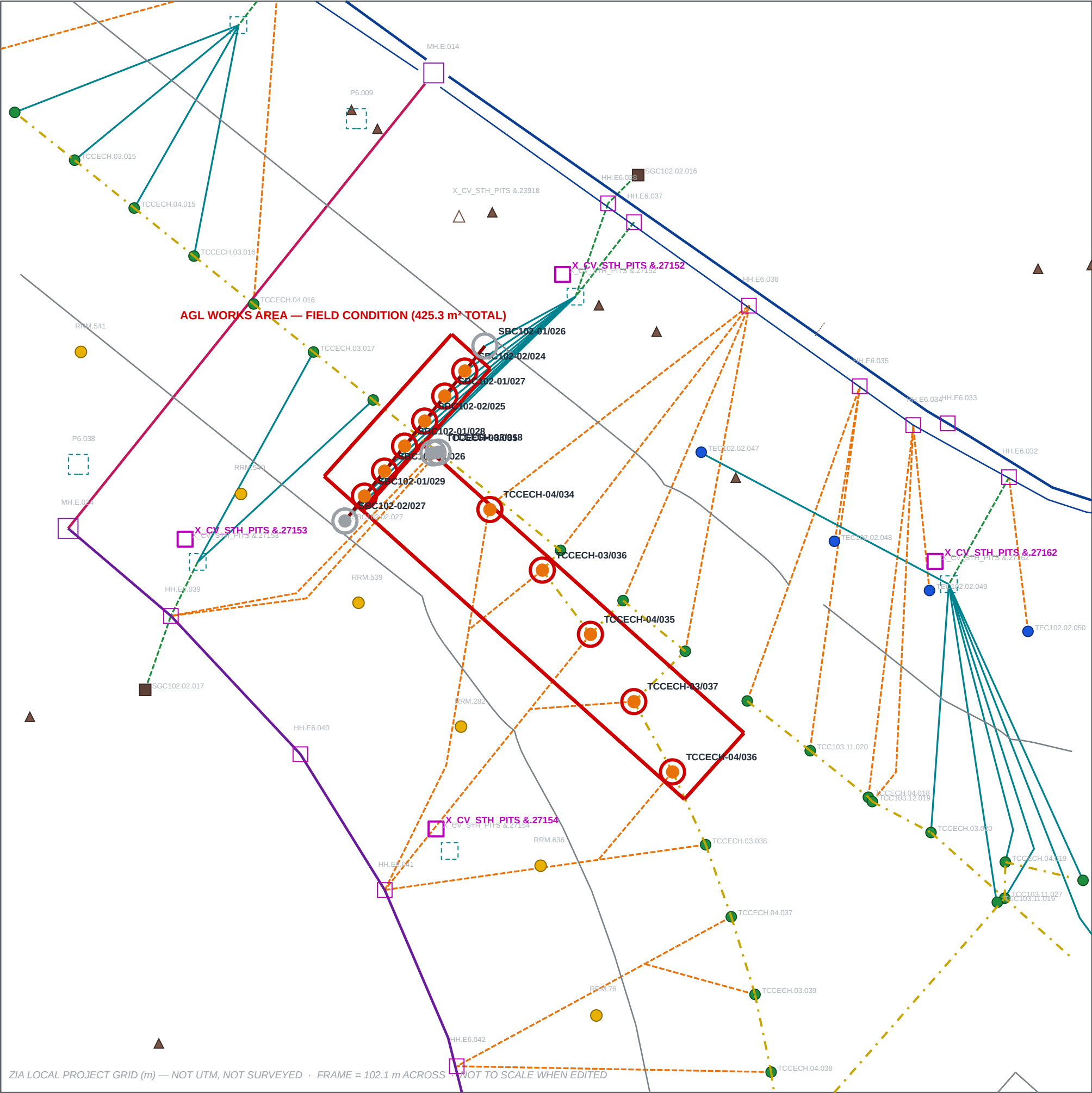
REV P08 — WHAT CHANGED

All secondary routes saw cut at all three locations; no duct route remains in the AGL scope.

All new bases are SIDE-ENTRY SHALLOW BASE — 8" core at LOC-01, 12" at LOC-02/03.

Affected fittings reduce from 30 to 16: LOC-01 11, LOC-02 1 (TCCECH-03/008), LOC-03 4 (TCC only).

Coring now applies at all three locations. Rev P06/P07 stated coring at Location 1 only.



LOCATION 1 — 1001 · REV P08 (FINAL SCOPE)

FINAL SCOPE OF WORK 30.07.2026 (supersedes the Document_3 quantities) — FIELD CONDITION GOVERNS

Frame 102.1 m across · ZIA local grid (m) · duct geometry from Z1-Z2-Z3-MTA_SEGMENTATION source dwg · Rev P08 · issued 30.07.2026 · Prepared: Mohammed, AGL Team Leader — ADB SAFEGATE

SCOPE / QUANTITIES (PER FIELD VERIFICATION)

FINAL SCOPE (REV P08) — ALL 11 FITTINGS CORED, ALL ROUTES SAW CUT
CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE: 11 No.

- SBC (6): SBC102-02/024, 01/027, 02/025, 01/028, 02/026, 01/029
- TCC (5): TCCECH-04/034, 03/036, 04/035, 03/037, 04/036

ROUTE: SAW CUT throughout — the 6 No. SBC carried as VIA DUCT on Rev P06/P07 are saw cut at P08.
Saw cut detail per sheet SAW CUT & SIDE-ENTRY SHALLOW BASE DETAIL — detail drawing to be issued.

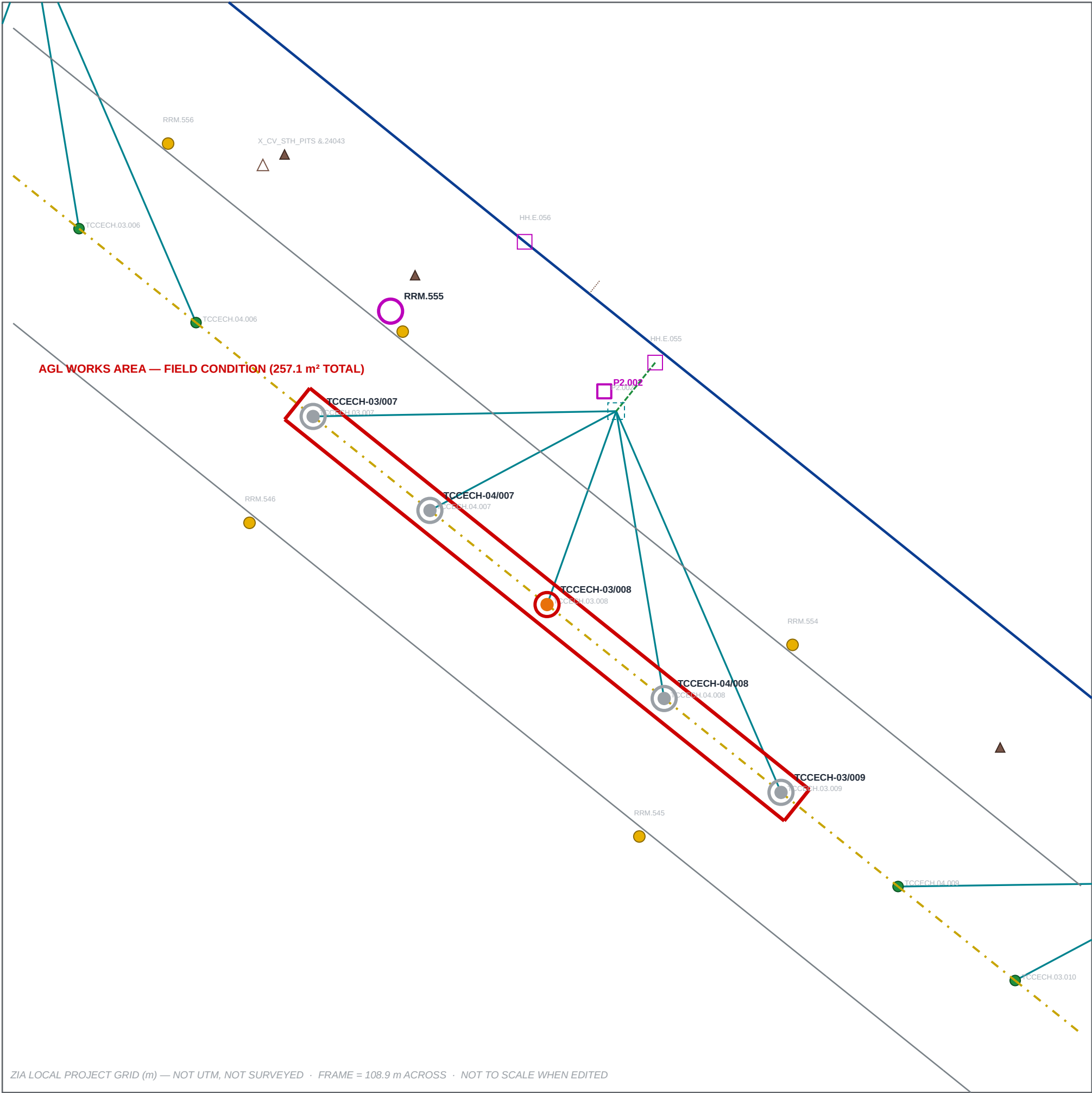
REMOVED FROM SCOPE AT REV P08 — no works: SBC102-02/027, TCCECH-03/035, TCCECH-03/018
FIELD VERIFIED NOT AFFECTED: SBC102-01/026 (no works)
TCC103 FITTINGS NOT IN FIELD SCOPE — SHOWN AS EXISTING ONLY
ISOLATE CIRCUITS: SBC102.01/.02, TCCECH.03/.04
SITE RECORD 23.07.2026: 9 cored + 2 reinstatement already worked
REV P08 SUPERSEDES the Rev P06/P07 quantities. SUPERSEDED = no works; NOT a field verification that the asset is unaffected.
RED AGL WORKS AREA = Rev P05 field-condition milling extent, UNCHANGED at P08 — the civil milling limit, not a hull of the affected assets.

GENERAL NOTES

- Positions plotted are ZIA LOCAL PROJECT GRID metres — NOT UTM 40N. The Rev P06 "EPSG:32640" label was wrong; no surveyed local-to-UTM transform exists. Do not issue a UTM coordinate from this sheet.
- AGL works area (red) per field condition — the Rev P05 milling extent, UNCHANGED at Rev P08. Coring at ALL THREE locations at P08: 8" at LOC-01, 12" at LOC-02 and LOC-03.
- Duct / ductbank / conduit / sawcut lines are REAL as-built geometry from ADA drawing Z1-Z2-Z3-MTA_SEGMENTATION.dxf — not light-to-nearest-pit construction lines. Layer name carries size and way-count; see sheet 6.
- Ducts are drawn as the separate segments they are in the source. Continuity NOT proven — do not take a continuous pull length off this sheet.
- Isolate, lock out & prove dead all listed circuits at CCR under permit before cutting, coring or excavation.
- HH / MH / transformer-pit / RRM symbols sit ~2.1-2.3 m off the source insertion point (sheet 6) — set out civil features from survey.
- Centerline & stop bar from as-built TCC / SBC fittings. Edge indicative 23 m — confirm on site.
- Every asset in frame carries its own as-built symbol, legended right. Type comes from the source layer, not the label. Assets with a works-action marker keep that marker.

LEGEND — THIS SHEET

WORKS ACTIONS & LINEWORK		AS-BUILT AGL ASSETS	
	CORE OUT 8" + SIDE-ENTRY BASE (SAWCUT)		TWY CENTRELINE LIGHT (31)
	SUPERSEDED — NOT IN REV P08 SCOPE		TWY EDGE LIGHT (4)
	FIELD VERIFIED — NOT AFFECTED		STOP BAR LIGHT (8)
	AS-BUILT DUCT / DUCTBANK 4×110 · 6×110		RUNWAY GUARD LIGHT / RRM (6)
	AS-BUILT SECONDARY CONDUIT		SIGN FOUNDATION (2)
	AGL WORKS AREA (GOVERNING)		HANDHOLE (HH) (11)
	TAXIWAY CENTRELINE (AS-BUILT)		TRANSFORMER HH (SEC. FEED HUB) (5)
	STOP BAR (AS-BUILT)		MANHOLE (MH) (2)
	TWY EDGE / PAVEMENT (INDICATIVE 23 m)		TRANSFORMER MH (SEC. FEED) (2)
			EARTHING POINT (10)
			EARTHING PIT (1)



ZIA LOCAL PROJECT GRID (m) — NOT UTM, NOT SURVEYED · FRAME = 108.9 m ACROSS · NOT TO SCALE WHEN EDITED

LOCATION 2 — 1002 · REV P08 (FINAL SCOPE)

FINAL SCOPE OF WORK 30.07.2026 (supersedes the Second_milling_area rev _1 quantities) — FIELD CONDITION GOVERNS

Frame 108.9 m across · ZIA local grid (m) · duct geometry from Z1-Z2-Z3-MTA_SEGMENTATION source dwg · Rev P08 · issued 30.07.2026 · Prepared: Mohammed, AGL Team Leader — ADB SAFEGATE

SCOPE / QUANTITIES (PER FIELD VERIFICATION)

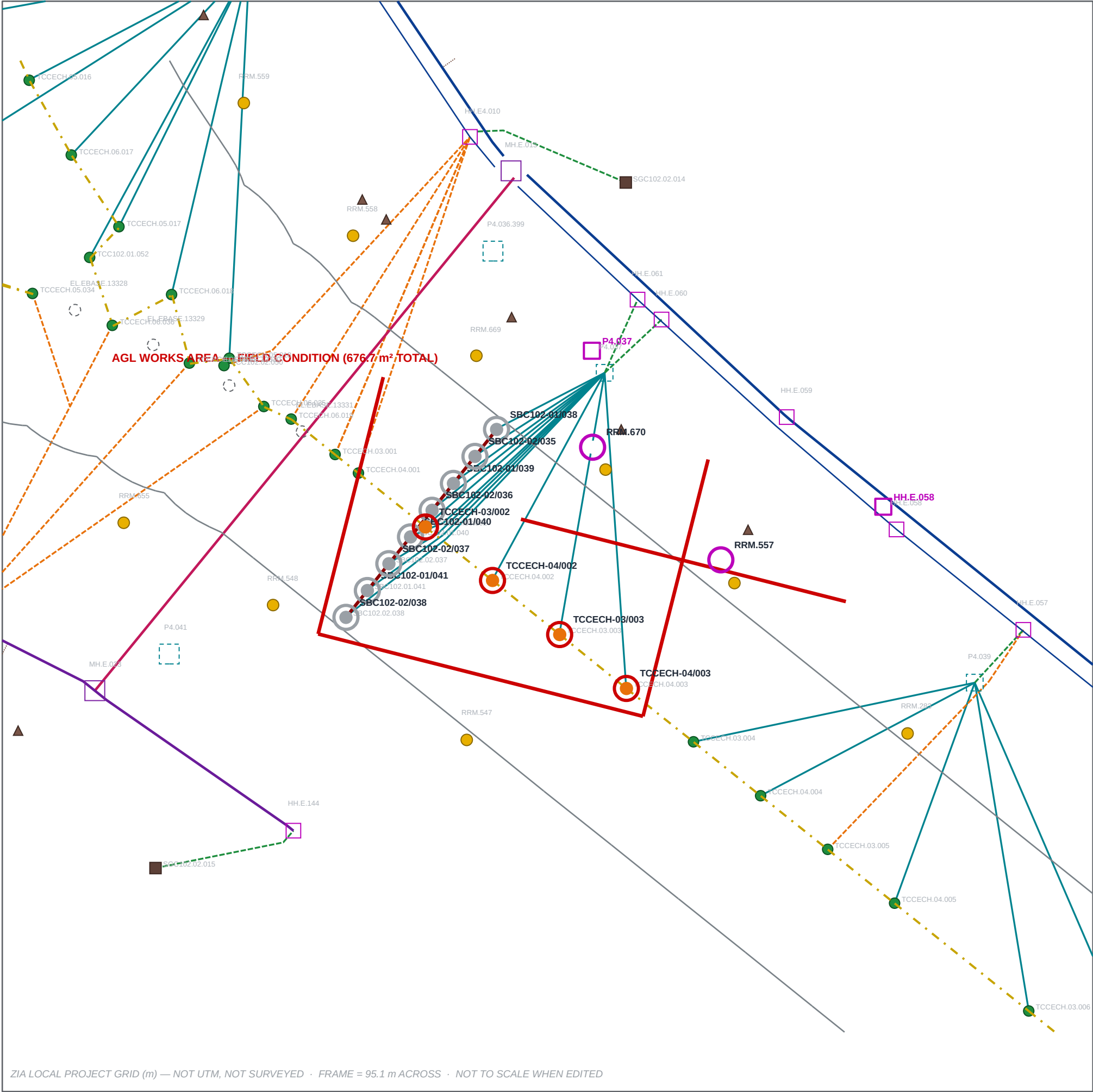
FINAL SCOPE (REV P08) — ONE FITTING AFFECTED, SAW CUT
CORE OUT EXISTING SHALLOW BASE + NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE: 1 No.
• TCCECH-03/008 — secondary route saw cut; existing shallow base cored out; new 12" side-entry base
Saw cut detail per sheet SAW CUT & SIDE-ENTRY SHALLOW BASE DETAIL — detail drawing to be issued.
REMOVED FROM SCOPE AT REV P08 — no works: TCCECH-03/007, 04/007, 04/008, 03/009
Rev P06/P07 carried all 5 No. as new-secondary-cable-only via duct. The final scope names 03/008 only.
RRM.555 (0.41 m from cut): REMOVE / PROTECT BEFORE SAWCUT, RE-FIX AFTER PAVING
ISOLATE CIRCUITS: TCCECH.03, TCCECH.04
REV P08 SUPERSEDES the Rev P06/P07 quantities. SUPERSEDED = no works; NOT a field verification that the asset is unaffected.
RED AGL WORKS AREA = Rev P05 field-condition milling extent, UNCHANGED at P08 — the civil milling limit, not a hull of the affected assets.

GENERAL NOTES

- Positions plotted are ZIA LOCAL PROJECT GRID metres — NOT UTM 40N. The Rev P06 "EPSG:32640" label was wrong; no surveyed local-to-UTM transform exists. Do not issue a UTM coordinate from this sheet.
- AGL works area (red) per field condition — the Rev P05 milling extent, UNCHANGED at Rev P08. Coring at ALL THREE locations at P08: 8" at LOC-01, 12" at LOC-02 and LOC-03.
- Duct / ductbank / conduit / sawcut lines are REAL as-built geometry from ADA drawing Z1-Z2-Z3-MTA_SEGMENTATION.dxf — not light-to-nearest-pit construction lines. Layer name carries size and way-count; see sheet 6.
- Ducts are drawn as the separate segments they are in the source. Continuity NOT proven — do not take a continuous pull length off this sheet.
- Isolate, lock out & prove dead all listed circuits at CCR under permit before cutting, coring or excavation.
- HH / MH / transformer-pit / RRM symbols sit ~2.1-2.3 m off the source insertion point (sheet 6) — set out civil features from survey.
- Centerline & stop bar from as-built TCC / SBC fittings. Edge indicative 23 m — confirm on site.
- Every asset in frame carries its own as-built symbol, legended right. Type comes from the source layer, not the label. Assets with a works-action marker keep that marker.

LEGEND — THIS SHEET

WORKS ACTIONS & LINEWORK		AS-BUILT AGL ASSETS	
	CORE OUT 12" + SIDE-ENTRY BASE (SAWCUT)		TWY CENTRELINE LIGHT (9)
	SUPERSEDED — NOT IN REV P08 SCOPE		RUNWAY GUARD LIGHT / RRM (5)
	RRM — REMOVE / PROTECT / RE-FIX		HANDHOLE (HH) (2)
	AS-BUILT DUCT / DUCTBANK 4x110 · 6x110		TRANSFORMER HH (SEC. FEED HUB) (1)
	AS-BUILT SECONDARY CONDUIT		EARTHING POINT (3)
	AGL WORKS AREA (GOVERNING)		EARTHING PIT (1)
	TAXIWAY CENTRELINE (AS-BUILT)		
	TWY EDGE / PAVEMENT (INDICATIVE 23 m)		



LOCATION 3 — 1003 · REV P08 (FINAL SCOPE)

FINAL SCOPE OF WORK 30.07.2026 (supersedes the Second_milling_area rev _1 quantities) — FIELD CONDITION GOVERNS

Frame 95.1 m across · ZIA local grid (m) · duct geometry from Z1-Z2-Z3-MTA_SEGMENTATION source dwg · Rev P08 · issued 30.07.2026 · Prepared: Mohammed, AGL Team Leader — ADB SAFEGATE

SCOPE / QUANTITIES (PER FIELD VERIFICATION)

FINAL SCOPE (REV P08) — 4 No. TCC ONLY, SAW CUT
NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE: 4 No.
• TCCECH-03/002, 04/002, 03/003, 04/003
ROUTE: SAW CUT — these 4 No. were carried as VIA DUCT on Rev P06/P07.
Saw cut detail per sheet SAW CUT & SIDE-ENTRY SHALLOW BASE DETAIL — detail drawing to be issued.
REMOVED FROM SCOPE AT REV P08 — no works, 8 No. SBC:
• SBC102-01/038, 01/039, 02/035, 02/036 (EP7 STOP BAR — was sawcut, base protect)
• SBC102-01/040, 01/041, 02/037, 02/038 (was new cable only, via duct)
RRM.557 (0.07 m — treat as within works) & RRM.670: REMOVE / PROTECT / RE-FIX
ISOLATE CIRCUITS: TCCECH.03/04 — SBC102.01/02 carries no AGL works here at P08;
confirm isolation against the saw cut alignment before works.
REV P08 SUPERSEDES the Rev P06/P07 quantities. SUPERSEDED = no works; NOT a field verification that the asset is unaffected.
RED AGL WORKS AREA = Rev P05 field-condition milling extent, UNCHANGED at P08 — the civil milling limit, not a hull of the affected assets.

GENERAL NOTES

- Positions plotted are ZIA LOCAL PROJECT GRID metres — NOT UTM 40N. The Rev P06 "EPSG:32640" label was wrong; no surveyed local-to-UTM transform exists. Do not issue a UTM coordinate from this sheet.
- AGL works area (red) per field condition — the Rev P05 milling extent, UNCHANGED at Rev P08. Coring at ALL THREE locations at P08: 8" at LOC-01, 12" at LOC-02 and LOC-03.
- Duct / ductbank / conduit / sawcut lines are REAL as-built geometry from ADA drawing Z1-Z2-Z3-MTA_SEGMENTATION.dxf — not light-to-nearest-pit construction lines. Layer name carries size and way-count; see sheet 6.
- Ducts are drawn as the separate segments they are in the source. Continuity NOT proven — do not take a continuous pull length off this sheet.
- Isolate, lock out & prove dead all listed circuits at CCR under permit before cutting, coring or excavation.
- HH / MH / transformer-pit / RRM symbols sit ~2.1-2.3 m off the source insertion point (sheet 6) — set out civil features from survey.
- Centerline & stop bar from as-built TCC / SBC fittings. Edge indicative 23 m — confirm on site.
- Every asset in frame carries its own as-built symbol, legended right. Type comes from the source layer, not the label. Assets with a works-action marker keep that marker.

LEGEND — THIS SHEET

WORKS ACTIONS & LINEWORK		AS-BUILT AGL ASSETS	
	CORE OUT 12" + SIDE-ENTRY BASE (SAWCUT)		TWY CENTRELINE LIGHT (23)
	SUPERSEDED — NOT IN REV P08 SCOPE		STOP BAR LIGHT (8)
	RRM — REMOVE / PROTECT / RE-FIX		RUNWAY GUARD LIGHT / RRM (9)
	AS-BUILT DUCT / DUCTBANK 4×110 · 6×110		EXISTING LIGHT BASE (NO FITTING) (4)
	AS-BUILT SECONDARY CONDUIT		SIGN FOUNDATION (2)
	AGL WORKS AREA (GOVERNING)		HANDHOLE (HH) (7)
	TAXIWAY CENTRELINE (AS-BUILT)		TRANSFORMER HH (SEC. FEED HUB) (2)
	STOP BAR (AS-BUILT)		MANHOLE (MH) (2)
	TWY EDGE / PAVEMENT (INDICATIVE 23 m)		TRANSFORMER MH (SEC. FEED) (2)
			EARTHING POINT (7)

CONSOLIDATED FIELD-GOVERNED SCOPE — TWY E (E4–E6) · REV P08 (FINAL SCOPE)

Loc	Asset	Works Action	Route	Remarks
LOC-01	SBC102-02/024	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-01/027	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-02/025	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-01/028	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-02/026	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-01/029	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	TCCECH-04/034	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	TCCECH-03/036	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	TCCECH-04/035	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	TCCECH-03/037	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	TCCECH-04/036	CORE OUT 8" + NEW SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	8" core out; new side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-01	SBC102-02/027	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-01	TCCECH-03/035	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-01	TCCECH-03/018	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-01	SBC102-01/026	NOT AFFECTED	—	Field verified — no works
LOC-02	TCCECH-03/008	CORE OUT EXISTING BASE + NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	Existing shallow base cored out; new 12" side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-02	TCCECH-03/007	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-02	TCCECH-04/007	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-02	TCCECH-04/008	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-02	TCCECH-03/009	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-02	RRM.555	RRM REMOVE / PROTECT	—	Remove / protect before saw cut, re-fix after paving
LOC-03	TCCECH-03/002	NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	New 12" side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-03	TCCECH-04/002	NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	New 12" side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-03	TCCECH-03/003	NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	New 12" side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-03	TCCECH-04/003	NEW 12" SIDE-ENTRY SHALLOW BASE + NEW CABLE	Sawcut	New 12" side-entry shallow base; full manhole-to-light cable replacement via saw cut
LOC-03	SBC102-01/038	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-01/039	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-02/035	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-02/036	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-01/040	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-01/041	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-02/037	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	SBC102-02/038	NOT IN REV P08 SCOPE	—	In the Rev P06/P07 scope; not named in the final scope of work 30.07.2026. No works. Not a field verification of 'not affected'
LOC-03	RRM.557	RRM REMOVE / PROTECT	—	Remove / protect before saw cut, re-fix after paving
LOC-03	RRM.670	RRM REMOVE / PROTECT	—	Remove / protect before saw cut, re-fix after paving

Totals (Rev P08 final scope): 16 fittings — 11 cored 8" at LOC-01, 5 cored 12" at LOC-02/03 — all with a new side-entry shallow base and a new secondary cable, all via SAW CUT. · 15 assets removed from scope at P08 (in the Rev P06/P07 scope, not in the final scope — no works). · 3 RRM's remove / protect / re-fix. · 1 field verified not affected. · No duct route remains in the AGL scope; the "Route" column reads Sawcut throughout by instruction, not by measurement off the as-built duct geometry.

SAW CUT & SIDE-ENTRY SHALLOW BASE DETAIL — TWY E (E4–E6) · REV P08 · HOLD

DETAIL — HELD FOR ISSUE

SAW CUT DETAIL DRAWING TO BE ISSUED

The final scope of work records that the saw cut detail drawing will be provided. It has not been issued at the date of this revision.

Paste the issued detail into this frame and re-issue the sheet. Until it is issued and accepted, NO SAW CUTTING IS TO COMMENCE — hold point H6.

SAW CUT & CORING SCHEDULE — REV P08 FINAL SCOPE					
Location	Fittings	Core	New base	Route	Sheet
LOC-01	11 (6 SBC · 5 TCC)	8"	Side-entry shallow base	Saw cut	1001
LOC-02	1 (TCCECH-03/008)	12"	Side-entry shallow base — existing base cored out	Saw cut	1002
LOC-03	4 (TCC only)	12"	Side-entry shallow base	Saw cut	1003
TOTAL	16	11 @ 8" · 5 @ 12"	16 side-entry shallow bases	Saw cut throughout	—

SIDE-ENTRY SHALLOW BASE — WHAT THE DETAIL MUST ALSO SHOW

10. Side-entry shallow base — depth, entry height above the base floor, entry bore and the seal at the entry.

11. Core diameter and core depth for 8" and for 12", and the grout / bedding specification for each.

12. At Location 2, the method for coring out the EXISTING shallow base and making good before the new 12" base is set.

13. Setting-out tolerance of the base relative to the as-built fitting position, and the level tolerance to finished pavement.

WHAT THE SAW CUT DETAIL MUST ANSWER BEFORE IT CAN BE BUILT TO

Cut geometry

1. Saw cut width and depth, and the tolerance on each.

2. Minimum cover from finished pavement surface to the top of the new secondary cable.

3. Cut termination at the light base — how the cut meets the side entry, and the radius or chamfer at the entry.

4. Minimum offset from slab joints, existing cuts and the milling edge, and what to do where the cut would run within that offset.

Cable and backfill

5. Cable protection within the cut — sleeve, tape or direct — and its fixing.

6. Backfill and sealant specification, layer by layer, with cure time before the area returns to operational service.

7. Reinstatement of the pavement surface and the finished surface tolerance.

Transition

8. How the saw cut route transitions into the existing duct in the NON-CONSTRUCTION area — this is open technical query Q3 and it governs the ends of every run.

9. Treatment where the saw cut crosses the existing 4 x 110 mm secondary duct bank exposed at 50 mm milling depth.

Items 8 and 9 are the two that are not purely detailing. Item 8 is open technical query Q3 — the transition into the non-construction area was never answered, and every run has two ends. Item 9 is a physical clash with a duct bank the field already found at 50 mm. Neither can be closed out on site.

SHEET 6 — DUCT PROVENANCE & REGISTRATION VERIFICATION · TWY E (E4-E6) REV P06

1 · SOURCE OF THE DUCT GEOMETRY

Drawing: Z1-Z2-Z3-MTA_SEGMENTATION.dxf, issued by Abu Dhabi Airports. AC1032 (AutoCAD 2018), \$INSUNITS = 6 (metres), 549 MB, 2,564 layers, 12 model-space xrefs all inserted at (0,0,0), scale 1, rotation 0.
Extracted population: 33,720 fixture INSERTs and 49,505 linear features (1,878 km), of which 17,101 features / 466 km are duct, ductbank, conduit, sawcut or secondary cable.
Duct type, bore and way-count are read from the layer name — that is the only place the drawing records them (e.g. CV_OUTER DUCT 4x110mm dia).
Extraction is reconciled against an independent per-layer entity census; every count above is asserted by an automated check, not read off by eye.

2 · HOW SHEETS 2-4 WERE TIED TO THE SOURCE DRAWING

The sheets carry no coordinate grid, so each was registered to the source drawing by matching the plotted AGL fitting pattern: type-matched 2-point RANSAC to find the correspondence, then a 4-parameter Helmert fit (translation, rotation, uniform scale) by least squares — not an affine, which would hide error as fake shear. Half the matched pairs were held back from the fit; the hold-out RMS below is the accuracy figure.

Sheet	Fitting symbols in frame	Matched to source within 50 mm	Fit RMS	HOLD-OUT RMS (pairs never fitted)	Plotted scale	Frame on ground
LOC-01	64	37	0.4 mm	0.40 mm on 14	93,614 EMU/m	102.1 m
LOC-02	18	9	0.3 mm	0.46 mm on 4	87,716 EMU/m	108.9 m
LOC-03	59	11	58.2 mm	0.49 mm on 13	100,476 EMU/m	95.1 m

Rotation solved to +0.0001° / −0.000001° / +0.0027° and the y axis is mirrored — i.e. the sheets are plotted axis-aligned to the grid.
Independent check: the deck's own "as-built taxiway centreline" polyline, never used in the fit, lands on source centreline fixtures at 0.3–0.4 mm median on LOC-01/02.

3 · FINDING — THE Rev P05 COORDINATE LABEL WAS WRONG

Rev P05 sheets state "UTM 40N (EPSG:32640) — AS-BUILT POSITIONS". They are not UTM. The plotted positions are ZIA local project grid metres (X ≈ 5,800–6,200, Y ≈ 54,400–54,700). UTM 40N here would read easting ~230,000–800,000 and northing ~2,650,000–2,750,000.
The source drawing carries no GEODATA object and no georeferencing of any kind, so no local→UTM transform can be derived from it. Rev P06 sheets are relabelled accordingly.
Consequence: any coordinate previously issued from Rev P05 as UTM — to ADA, Injaz or a contractor — needs withdrawing and re-checking against survey control.

5 · LIMITS — READ BEFORE USING THIS GEOMETRY

- a) Runs are fragments, not proven pulls. The duct lines are the individual LINE / LWPOLYLINE segments as drafted. No endpoint snapping or graph chaining has been done, so "handhole A to handhole B" is not established. Do not order cable to a length scaled off these sheets.
- b) Coordinates are ZIA local grid, not surveyed and not UTM. Nothing here may be issued as a setting-out coordinate without an externally checked transform against ADA survey control.
- c) Civil symbols are ~2.1–2.3 m off. Handhole, manhole, transformer-pit and RRM symbols on the sheets sit a mean 2.1–2.3 m (sd 0.3–0.45 m) from the source-drawing insertion point, while AGL light fittings match to sub-millimetre. Set out civil features from survey, not from these sheets.
- d) Per-asset route attribution is NOT given. The consolidated table's "Route" column is unchanged from Rev P05. Attributing a specific duct layer to a named asset was attempted and rejected: the sheets' labels sit 3–8 m from their markers with neighbours only 10–15% further, and several markers resolve to the same label, so the assignment cannot be made from the drawing alone. It needs the source asset register, which is not in this data set.
- e) Z values are unreliable in the source drawing. This is a 2D basemap only; take levels from survey.
- f) Elements outside the twelve identity-inserted xrefs were excluded rather than drawn in the wrong place (3,375 geometry entities, 2,409 nested fixtures drawing-wide). If a duct looks absent in a given corner, check this first.

4 · DUCT SCHEDULE — WHAT IS DRAWN ON EACH SHEET

LOC-01 — 59 segments, 1433 m inside the frame

Source layer (carries bore & way-count)	Draw	Segs	On sheet
CV-NEW SEC SAWCUT		23	606.7 m
CV-EX SEC CONDUIT		21	490.9 m
CV_INNER DUCT 4x110mm dia		3	84.6 m
CV_OUTER DUCT 4x110mm dia		2	81.5 m
CV_STH DUCT 6x110mm dia		1	65.6 m
CV_DUCT CROSSING 6x110mm dia		1	53.3 m
CV-NEW SEC CONDUIT		7	49.0 m
CV_GRND DUCT		1	1.5 m

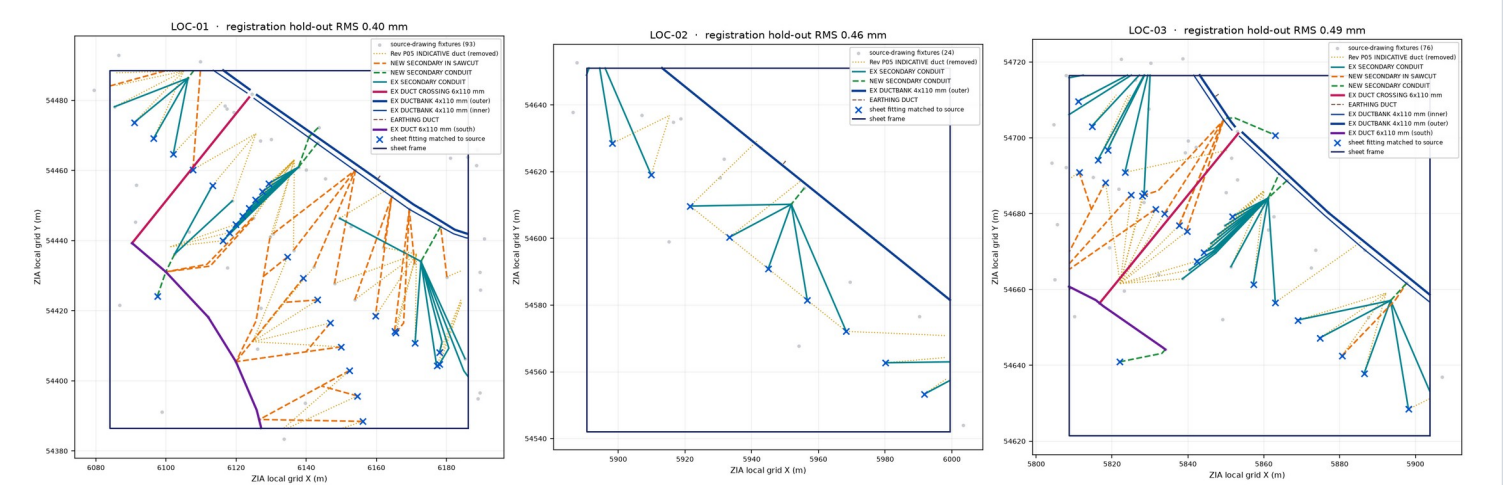
LOC-02 — 13 segments, 349 m inside the frame

Source layer (carries bore & way-count)	Draw	Segs	On sheet
CV-EX SEC CONDUIT		10	230.7 m
CV_OUTER DUCT 4x110mm dia		1	110.8 m
CV-NEW SEC CONDUIT		1	6.2 m
CV_GRND DUCT		1	1.6 m

LOC-03 — 53 segments, 1104 m inside the frame

Source layer (carries bore & way-count)	Draw	Segs	On sheet
CV-EX SEC CONDUIT		29	541.5 m
CV-NEW SEC SAWCUT		11	259.1 m
CV_INNER DUCT 4x110mm dia		2	84.2 m
CV_OUTER DUCT 4x110mm dia		2	81.7 m
CV_DUCT CROSSING 6x110mm dia		1	57.8 m
CV-NEW SEC CONDUIT		5	46.6 m
CV_STH DUCT 6x110mm dia		1	30.4 m
CV_GRND DUCT		2	2.4 m

6 · THE GEOMETRY, IN THE SOURCE DRAWING'S OWN GRID



SHEET 7 — SECONDARY FEED TOPOLOGY (DERIVED) · STAR FROM TRANSFORMER HANDHOLES, NOT A CHAINED RUN

FINDING — THE Rev P05 LOC-02 SCOPE NOTE DESCRIBES THE WRONG TOPOLOGY

Rev P05, sheet 3 (LOC-02) states: "SECONDARY DUCT: CHAINED RUN HH.E.056 → 03/007 → 04/007 → 03/008 → 04/008 → 03/009 → HH.E.054". The source drawing shows no such chain. Every affected light at LOC-02 is fed by its own individual conduit spur radiating from ONE transformer handhole — five separate home runs of 20.5, 21.1, 29.1, 30.3 and 41.5 m, 142.5 m in total, not a series loop between lights. This changes the cable quantity, the number of pulls and the isolation sequence. Re-price and re-plan LOC-02 before ordering cable.

WHY THIS ALSO PROVES THE GEOMETRY IS REAL

The conduit fragments were chained by snapping endpoints within 0.30 m. What comes out is not an arbitrary set of lines — it is a working secondary distribution network:

- Every hub node lands on a CV_ETRANS HH (existing transformer handhole) fixture at 0.000-0.002 m. Four hubs at LOC-01, one at LOC-02, two at LOC-03.
- 40 of 45 spurs terminating at a light land within 50 mm of that light's insertion point; most are exactly 0.00 m.
- Spur lengths are 6–42 m, consistent with transformer-handhole-to-light home runs.

Straight light-to-nearest-pit construction lines cannot produce that. Invented or inferred geometry would not terminate on fixture insertion points to the millimetre, nor converge specifically on transformer handholes rather than on any nearby pit.

METHOD AND ITS LIMIT

The source drawing stores these conduits as loose LINE / LWPOLYLINE segments with no connectivity. The chaining here is DERIVED, not read from the drawing: endpoints within 0.30 m were treated as one node, then each branch was walked outward from its hub until it reached a leaf or a junction. Consequence: a spur length is the drawn geometry of a branch, not a proven pull. Two conduits passing within 0.30 m without physically joining would be merged; a genuine joint drawn with a wider gap would be split. Prove each run on site before ordering. The 0.30 m tolerance sits far below the shortest real spur (6.2 m) and well above the drafting gaps observed. No spur changed hop count when the tolerance was varied between 0.10 and 0.50 m.

LOC-01 — DERIVED SPUR SCHEDULE			
4 transformer-handhole hub(s) · 26 spurs · 516.9 m to lights			
Length	Terminates at	Source layer	Offset
35.44 m	Taxiway centreline light	EL-TAXICL	5.13 m !
31.36 m	Taxiway centreline light	EL-TAXICL	0.00 m
30.40 m	Taxiway centreline light	EL-TAXICL	0.00 m
30.08 m	Taxiway centreline light	EL-TAXICL	0.06 m !
30.08 m	Stop bar light	EL-STOPBAR	0.04 m
27.17 m	Stop bar light	EL-STOPBAR	0.00 m
26.84 m	Taxiway centreline light	EL-TAXICL	0.00 m
26.21 m	Taxiway edge light	EL-TAXIWAY EDGE	0.07 m !
26.02 m	Taxiway edge light	EL-TAXIWAY EDGE	0.00 m
24.20 m	Stop bar light	EL-STOPBAR	0.00 m
23.28 m	Taxiway centreline light	EL-TAXICL	0.00 m
22.49 m	Taxiway centreline light	EL-TAXICL	0.03 m
22.42 m	Taxiway centreline light	EL-TAXICL	0.00 m
22.33 m	Taxiway centreline light	EL-TAXICL	0.00 m
21.99 m	Taxiway centreline light	EL-TAXICL	0.00 m
21.24 m	Stop bar light	EL-STOPBAR	0.00 m
19.86 m	Taxiway centreline light	EL-TAXICL	0.00 m
19.69 m	Taxiway centreline light	EL-TAXICL	0.00 m
18.29 m	Stop bar light	EL-STOPBAR	0.00 m
15.37 m	Stop bar light	EL-STOPBAR	0.00 m
12.96 m	Sign foundation	EL-SGN-FNDTN	0.00 m
12.84 m	Sign foundation	EL-SGN-FNDTN	0.28 m !
12.48 m	Stop bar light	EL-STOPBAR	0.00 m
9.66 m	Stop bar light	EL-STOPBAR	0.00 m
8.85 m	Handhole	CV_HH	0.00 m
2.83 m	Existing transformer handhole	CV_ETRANS HH	2.83 m !
! offset >50 mm — confirm the termination on site			

LOC-02 — DERIVED SPUR SCHEDULE			
1 transformer-handhole hub(s) · 6 spurs · 142.5 m to lights			
Length	Terminates at	Source layer	Offset
41.50 m	Taxiway centreline light	EL-TAXICL	0.00 m
30.30 m	Taxiway centreline light	EL-TAXICL	0.00 m
29.10 m	Taxiway centreline light	EL-TAXICL	0.00 m
21.10 m	Taxiway centreline light	EL-TAXICL	0.00 m
20.52 m	Taxiway centreline light	EL-TAXICL	0.00 m
6.23 m	Handhole	CV_HH	0.00 m
! offset >50 mm — confirm the termination on site			

LOC-03 — DERIVED SPUR SCHEDULE			
2 transformer-handhole hub(s) · 20 spurs · 412.1 m to lights			
Length	Terminates at	Source layer	Offset
31.97 m	Taxiway centreline light	EL-TAXICL	0.00 m
31.13 m	Stop bar light	EL-STOPBAR	0.00 m
29.02 m	Taxiway centreline light	EL-TAXICL	0.00 m
28.16 m	Stop bar light	EL-STOPBAR	0.00 m
27.57 m	Taxiway centreline light	EL-TAXICL	0.00 m
25.99 m	Runway guard light / RRM	EL-RRM	4.99 m !
25.23 m	Stop bar light	EL-STOPBAR	0.00 m
25.07 m	Taxiway centreline light	EL-TAXICL	0.00 m
23.13 m	Taxiway centreline light	EL-TAXICL	0.00 m
22.34 m	Stop bar light	EL-STOPBAR	0.00 m
21.12 m	Taxiway centreline light	EL-TAXICL	0.00 m
20.68 m	Taxiway centreline light	EL-TAXICL	0.07 m !
20.57 m	Taxiway centreline light	EL-TAXICL	0.00 m
20.47 m	Taxiway centreline light	EL-TAXICL	0.00 m
19.24 m	Stop bar light	EL-STOPBAR	0.00 m
16.34 m	Stop bar light	EL-STOPBAR	0.00 m
13.46 m	Stop bar light	EL-STOPBAR	0.00 m
10.65 m	Stop bar light	EL-STOPBAR	0.00 m
7.00 m	Handhole	CV_HH	0.00 m
6.80 m	Handhole	CV_HH	0.00 m
! offset >50 mm — confirm the termination on site			

SHEET 8 — AS-BUILT AGL ASSET SYMBOL SCHEDULE · TWY E (E4-E6) REV P06

HOW TO READ THIS

Circle = light · square = civil chamber · triangle = earthing · filled rectangle = sign foundation. Lights are coloured to the light's own showing: taxiway centreline green, taxiway edge blue, stop bar red, runway guard amber. A dashed outline marks the source drawing's "existing" layer for that family. Asset type is taken from the source-drawing LAYER, never from the label on the sheet. Assets that carry a Rev P05 works-action marker keep that marker instead of a type symbol — their type is in the ID label and in this schedule. Counts are assets inside each sheet frame.

SYMBOL · ASSET TYPE · SOURCE LAYER · COUNT PER SHEET

Symbol	Asset type	Source drawing layer(s)	LOC-01	LOC-02	LOC-03	Total in frame
	TWY CENTRELINE LIGHT	EL-TAXICL	31	9	23	63
	TWY EDGE LIGHT	EL-TAXIWAY EDGE	4	—	—	4
	STOP BAR LIGHT	EL-STOPBAR	8	—	8	16
	RUNWAY GUARD LIGHT / RRM	EL-RRM	6	5	9	20
	EXISTING LIGHT BASE (NO FITTING)	EL-EBASE	—	—	4	4
	SIGN FOUNDATION	EL-SGN-FNDTN	2	—	2	4
	HANDHOLE (HH)	CV_HH	11	2	7	20
	TRANSFORMER HANDHOLE (SECONDARY FEED HUB)	CV_ETRANS HH	5	1	2	8
	MANHOLE (MH)	CV_MH	2	—	2	4
	TRANSFORMER MANHOLE (SEC. FEED)	CV_ETRANS MH	2	—	2	4
	EARTHING POINT	CV_GRND	10	3	7	20
	EARTHING PIT	CV_GRND PIT	1	1	—	2
TOTAL — 12 asset types			82	21	66	169

Of these, Rev P05 drew no symbol at all for 42 assets on LOC-01, 12 on LOC-02 and 31 on LOC-03 — they are added in Rev P06 from the as-built and are NOT field-verified. asset_census.json accompanies this deck.